Priority-Aware Task Offloading and 3D Trajectory Optimization in NOMA-Enabled Multi-UAV MEC-IoT Networks Using MAPPO

Nayanjit Talukdar, Kushagra Gupta, Krishna Sikheriya, Aditya Pankaj Sharma, Nabajyoti Mazumdar, *Senior Member, IEEE*

***Abstract*—Unmanned aerial vehicle (UAV)-assisted mobile edge computing (MEC) has emerged as a promising paradigm for providing low-latency computation services to resource-constrained Internet of Things (IoT) devices in dynamic environments. However, existing task offloading frameworks largely treat all tasks homogeneously, ignoring the heterogeneous urgency and importance levels inherent in real-world IoT applications. Moreover, jointly optimizing priority-aware task offloading, NOMA-based uplink communication, and 3D multi-UAV trajectory control under UAV energy constraints remains a challenging open problem due to the hybrid discrete-continuous decision space and the tight coupling among these components. To address this, we propose a *Priority-Aware Multi-UAV Joint Optimization (PA-MUJO)* framework that formulates the problem as a stochastic mixed-integer nonlinear program (MINLP) and solves it via a multi-agent proximal policy optimization (MAPPO) algorithm under the centralized training with decentralized execution (CTDE) paradigm. The proposed system introduces a composite task priority model that captures both deadline-based urgency and application-type importance, enabling the UAV fleet to preferentially serve high-priority tasks. NOMA-based uplink transmission powers are computed analytically via bisection, decoupling communication optimization from the reinforcement learning (RL) policy and reducing action-space dimensionality. Simulation results demonstrate that PA-MUJO significantly reduces priority-weighted task completion latency and UAV energy consumption compared to state-of-the-art baselines, while maintaining effective multi-UAV coordination.**

***Index Terms*—Unmanned aerial vehicles, mobile edge computing, Internet of Things, multi-agent proximal policy optimization, NOMA, priority-aware task offloading, 3D trajectory optimization.**

I. Introduction

THE rapid proliferation of Internet of Things (IoT) devices, driven by emerging applications such as smart cities, industrial automation, environmental monitoring, and disaster response, demands ultra-reliable and low-latency computation services [1], [2]. However, the limited computation capacity, energy budget, and storage of IoT devices (hereafter referred to as wireless devices, WDs) make local execution of increasingly complex and data-intensive tasks infeasible. Unmanned aerial vehicles (UAVs) equipped with mobile edge computing (MEC) servers have emerged as a compelling solution due to their flexible deployment, line-of-sight (LoS)-dominant air-to-ground (A2G) channels, and ability to provide on-demand computation services in dynamic environments [3]–[5].

Despite growing interest in UAV-assisted MEC, three critical gaps persist in the literature. *First*, the majority of existing task offloading frameworks [6]–[8] treat all tasks homogeneously, ignoring the heterogeneous urgency and importance levels that naturally arise in real-world IoT deployments. In practice, tasks generated by different IoT applications exhibit vastly different latency sensitivities and criticality levels. For instance, a fire-detection alert in a smart-city surveillance system requires far more urgent processing than a routine environmental sensor reading. Failing to account for such priority differences leads to suboptimal resource utilization, where high-priority tasks may experience unacceptable delays while low-priority tasks consume disproportionate computation and communication resources.

Second, while multi-agent reinforcement learning (MARL) frameworks such as [4] have been applied to multi-UAV trajectory and offloading optimization, they often employ independent or loosely coupled agent architectures that struggle to scale in tightly coupled decision spaces. In particular, UAV trajectory, task offloading, and NOMA-based uplink communication are inherently interdependent: UAV mobility affects channel conditions and coverage; offloading decisions reshape traffic patterns and interference; and NOMA power allocation determines achievable uplink rates. This tight coupling creates a combinatorial optimization challenge that existing single-agent or loosely coordinated multi-agent approaches cannot tractably address.

Third, most existing works optimize UAV trajectories in 2D [7], neglecting the significant impact of altitude control on A2G channel quality, interference management, and propulsion energy consumption. Recent studies [9], [10] have shown that altitude adaptation can substantially improve communication performance and energy efficiency, yet integrating 3D trajectory optimization with priority-aware offloading and NOMA communication within a unified learning framework remains largely unexplored.

To address these limitations, we propose the *Priority-Aware Multi-UAV Joint Optimization (PA-MUJO)* framework. PA-MUJO introduces a composite task priority model that jointly captures deadline-based urgency and application-type impor-

(Corresponding author: Nabajyoti Mazumdar.)

K. Gupta (student ID: IIT2023133), K. Sikheriya (student ID: IIT2023139), and A. P. Sharma (student ID: IIT2023162) are B.Tech. students. N. Talukdar is a Ph.D. Scholar. All four are with the Department of Information Technology, Indian Institute of Information Technology, Allahabad, India. N. Mazumdar is an Assistant Professor with the same department. (e-mail: nabajyoti@iiita.ac.in).

tance weights, enabling the multi-UAV system to preferentially allocate resources to high-priority tasks. The optimization problem is formulated as a stochastic mixed-integer nonlinear program (MINLP) that jointly optimizes 3D UAV trajectories, priority-aware task offloading decisions, and NOMA-based uplink power allocation, subject to UAV energy, mobility, and quality-of-service (QoS) constraints. The problem is solved via multi-agent proximal policy optimization (MAPPO) under the centralized training with decentralized execution (CTDE) paradigm, where a centralized critic guides policy learning during training while each UAV makes real-time decisions using only local observations during execution. NOMA transmit powers are computed analytically through bisection-based optimization, decoupling communication from the RL action space and improving learning stability.

The main contributions of this paper are as follows:

- We formulate a priority-aware multi-UAV MEC-IoT optimization problem as a stochastic MINLP that jointly models heterogeneous task priorities, 3D UAV trajectory control, NOMA-based uplink communication, and UAV energy constraints.
- We propose a composite task priority model that integrates deadline-based urgency with application-type importance weights, enabling differentiated service for IoT tasks with heterogeneous criticality levels.
- We develop a MAPPO-based multi-agent learning framework under the CTDE paradigm for joint 3D trajectory optimization and priority-aware task offloading, where analytical NOMA power allocation via bisection decouples communication from the RL policy to reduce action dimensionality and improve training stability.
- We conduct extensive simulations to validate that PA-MUJO significantly outperforms benchmark methods in priority-weighted task completion latency and UAV energy consumption under varying network conditions.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system model and problem formulation. Section IV describes the proposed methodology. Section V presents simulation results, and Section VI concludes the paper.

II. Related Work

A. Task Offloading in UAV-Enabled MEC

Task offloading in UAV-enabled MEC has been extensively studied to reduce latency and energy consumption. Early approaches employed convex optimization and game-theoretic methods for joint offloading and resource allocation [8], while recent studies have leveraged deep reinforcement learning (DRL) to capture dynamic system interactions [6], [11]. Nguyen et al. [12] studied dependency-aware task offloading in collaborative UAV-MEC networks, using a discrete whale optimization algorithm with convex resource allocation to minimize latency for DAG-based dependent subtasks under UAV energy constraints. Multi-agent learning frameworks have further enabled distributed decision-making across UAVs [4], [7]. However, these approaches focus primarily on offloading efficiency in isolation, treating all tasks homogeneously without considering the heterogeneous priority levels inherent in real-world IoT deployments.

B. Priority-Aware Task Offloading

Priority-aware offloading has received limited attention in the UAV-MEC literature. Hu et al. [13] proposed a priority-based offloading scheme for single-UAV MEC, where tasks are ranked by urgency and served accordingly, but the approach does not extend to multi-UAV coordination or NOMA-based communication. Sun et al. [14] introduced a priority-based resource allocation framework that considers task deadlines, but their method relies on heuristic scheduling without learning-based adaptation to dynamic environments. Peng et al. [15] studied priority-aware multi-UAV scheduling using optimization techniques, but their trajectory design is limited to 2D and does not incorporate NOMA or multi-agent learning. In contrast, the proposed PA-MUJO framework integrates a composite priority model with MAPPO-based multi-agent learning, 3D trajectory optimization, and analytical NOMA power allocation within a unified framework.

C. Multi-UAV Trajectory Optimization

UAV trajectory optimization has been widely studied using both optimization-based and learning-based approaches. Two-timescale frameworks such as [16] decouple slow and fast-varying decisions to reduce complexity, but rely on heuristic approximations that struggle to scale to tightly coupled decision spaces. Zhan et al. [17] employed graph attention networks for joint UAV placement and dependent task offloading. Kang et al. [4] proposed a MAPPO-based approach for hierarchical aerial computing, demonstrating the effectiveness of multi-agent PPO for cooperative UAV resource allocation. However, most existing trajectory designs are limited to 2D horizontal movement, ignoring the benefits of altitude adaptation for A2G channel quality and propulsion energy management. Moreover, none of these works integrate priority-aware task offloading with 3D multi-UAV trajectory control under NOMA-based communication.

D. NOMA-Based Communication in UAV Networks

Power-domain NOMA has been widely adopted in UAV-assisted networks to improve spectral efficiency [3], [18]. Joint trajectory and NOMA resource allocation have been studied for single-UAV and multi-UAV settings, where successive interference cancellation (SIC) enables simultaneous uplink transmission from multiple WDs. However, existing NOMA-based UAV-MEC works typically optimize power allocation jointly with trajectory through the RL action space, increasing dimensionality and training complexity. In contrast, our framework analytically solves the NOMA power allocation subproblem via bisection, reducing the RL action dimensionality and improving convergence stability.

III. System Model and Problem Formulation

A. Network Model

As illustrated in Fig. 1, we consider a multi-UAV-enabled MEC-assisted IoT network operating over a finite mission

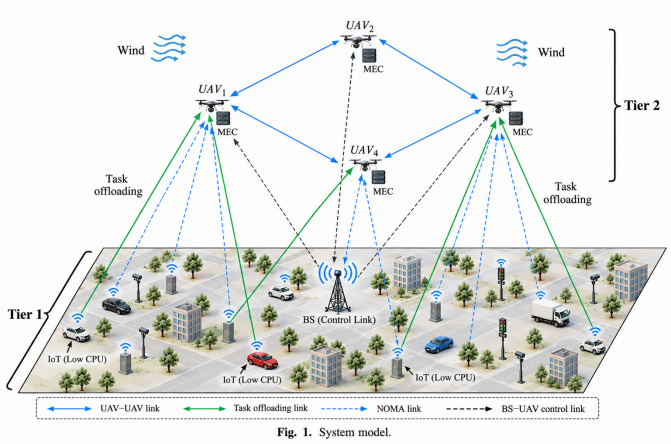


Fig. 1. System model.

Figure 1. Proposed multi-UAV MEC-IoT network architecture with priority-aware task offloading, NOMA-based uplink communication, and BS control link.

horizon divided into K decision epochs, indexed by $k \in \mathcal{K} \triangleq \{1, 2, \ldots, K\}$. The network consists of a set of ground-based WDs, a set of MEC-enabled UAVs, and one terrestrial base station (BS).

Let $\mathcal{N} \triangleq \{1, 2, \ldots, N\}$ denote the set of WDs. The position of WD $n \in \mathcal{N}$ at epoch k is denoted by $\mathbf{w}_n(k) = [x_n(k), y_n(k), 0]^\top$. Each WD has a local CPU frequency f_n^{loc} and may either execute a generated task locally or offload it to a serving UAV.

Let $\mathcal{M} \triangleq \{1, 2, \ldots, M\}$ denote the set of UAVs. UAV $m \in \mathcal{M}$ flies in 3D with position

$$\mathbf{q}_m(k) = [x_m(k), y_m(k), h_m(k)]^\top, \quad h_m(k) \in [h_{\min}, h_{\max}], \tag{1}$$

where $h_{\min}$ and $h_{\max}$ are the minimum and maximum allowable flight altitudes, respectively. Each UAV is equipped with an onboard MEC server of CPU frequency f_m^{uav}. UAVs execute offloaded tasks from WDs within their communication coverage.

A single BS is deployed at $\mathbf{b} = [x_b, y_b, 0]^\top$ and serves as a control anchor for the UAV fleet. The BS coordinates UAV deployment and provides network synchronization but does not participate in task computation. The system evolves in epochs of duration Δt, and the mobility of WDs and UAVs is captured through $\{\mathbf{w}_n(k)\}$ and $\{\mathbf{q}_m(k)\}$.

B. Mobility Model of WDs

At the beginning of the mission horizon, all WDs are uniformly distributed within the monitoring region $\mathcal{A}$. The mobility of each WD $n \in \mathcal{N}$ is modeled using the Gauss-Markov (GM) model [19], which captures temporal correlation in both speed and heading direction. At each epoch $k \in \mathcal{K}$, the speed $v_n(k)$ and heading angle $\theta_n(k)$ evolve as

$$v_n(k) = \rho_v v_n(k-1) + (1-\rho_v)\bar{v}_n + \sqrt{1-\rho_v^2}\,\xi_n^v(k), \tag{2}$$

$$\theta_n(k) = \rho_\theta \theta_n(k-1) + (1-\rho_\theta)\bar{\theta}_n + \sqrt{1-\rho_\theta^2}\,\xi_n^\theta(k), \tag{3}$$

where $\bar{v}_n$ and $\bar{\theta}_n$ denote the mean speed and heading direction of WD n, respectively. The parameters $\rho_v, \rho_\theta \in [0, 1]$ control the memory of the mobility process, while $\xi_n^v(k) \sim \mathcal{N}(0, \sigma_v^2)$ and $\xi_n^\theta(k) \sim \mathcal{N}(0, \sigma_\theta^2)$ are independent Gaussian perturbations. Based on the speed and heading at epoch $k-1$, the horizontal position of WD n is updated as

$$x_n(k) = x_n(k-1) + \Delta t\, v_n(k-1)\cos\theta_n(k-1), \tag{4}$$

$$y_n(k) = y_n(k-1) + \Delta t\, v_n(k-1)\sin\theta_n(k-1), \tag{5}$$

such that $\mathbf{w}_n(k) = [x_n(k), y_n(k), 0]^\top$. If the updated position moves outside $\mathcal{A}$, a specular reflection rule is applied to the heading direction to keep the WD trajectory within the monitoring region.

C. Communication Model

1) Air-to-Ground Channel Model: The primary offloading links between WDs and UAVs are modeled as probabilistic air-to-ground (A2G) channels, wherein the instantaneous propagation condition is either line-of-sight (LoS) or non-line-of-sight (NLoS) depending on the relative geometry and the surrounding environment [9]. For WD $n \in \mathcal{N}$ and UAV $m \in \mathcal{M}$ at epoch k, the Euclidean distance is given by

$$\begin{aligned} d_{n,m}(k) &= \|\mathbf{q}_m(k) - \mathbf{w}_n(k)\|_2 \\ &= \sqrt{\big(x_m(k) - x_n(k)\big)^2 + \big(y_m(k) - y_n(k)\big)^2 + h_m^2(k)}, \end{aligned} \tag{6}$$

and the corresponding elevation angle (in degrees) is

$$\phi_{n,m}(k) = \frac{180}{\pi} \arcsin\left(\frac{h_m(k)}{d_{n,m}(k)}\right). \tag{7}$$

The probability that the link between WD n and UAV m is LoS at epoch k is expressed as

$$\mathcal{P}_{n,m}^{\text{LoS}}(k) = \frac{1}{1 + a\exp\big(-b\big(\phi_{n,m}(k) - a\big)\big)}, \tag{8}$$

where a and b are environment-dependent constants. The NLoS probability is $\mathcal{P}_{n,m}^{\text{NLoS}}(k) = 1 - \mathcal{P}_{n,m}^{\text{LoS}}(k)$.

Let β_{LoS} and β_{NLoS} denote the large-scale channel gain coefficients under LoS and NLoS propagation, respectively, where $\beta_{\text{LoS}} > \beta_{\text{NLoS}} > 0$. The effective A2G channel gain between WD n and UAV m at epoch k is

$$\begin{aligned} g_{n,m}(k) &= \mathcal{P}_{n,m}^{\text{LoS}}(k)\, g_{n,m}^{\text{LoS}}(k) + \mathcal{P}_{n,m}^{\text{NLoS}}(k)\, g_{n,m}^{\text{NLoS}}(k) \\ &= \Big[\mathcal{P}_{n,m}^{\text{LoS}}(k)\beta_{\text{LoS}} + \mathcal{P}_{n,m}^{\text{NLoS}}(k)\beta_{\text{NLoS}}\Big]\big(d_{n,m}(k)\big)^{-\ell}, \end{aligned} \tag{9}$$

where ℓ denotes the path-loss exponent and $g_{n,m}^{\text{LoS}}(k) = \beta_{\text{LoS}}(d_{n,m}(k))^{-\ell}$, $g_{n,m}^{\text{NLoS}}(k) = \beta_{\text{NLoS}}(d_{n,m}(k))^{-\ell}$.

2) NOMA-Based Uplink Transmission: To enable simultaneous task offloading from multiple WDs to the same UAV, power-domain NOMA is adopted for the WD-to-UAV uplink [3]. Let $a_{n,m}(k) \in \{0, 1\}$ denote the association indicator, where $a_{n,m}(k) = 1$ means that WD n offloads its task to UAV m at epoch k, and $a_{n,m}(k) = 0$ otherwise. Each WD can associate with at most one UAV at each epoch:

$$\sum_{m \in \mathcal{M}} a_{n,m}(k) \le 1, \quad \forall n \in \mathcal{N},\ \forall k \in \mathcal{K}. \tag{10}$$

Let $\mathcal{N}_m(k) \triangleq \{n \in \mathcal{N} : a_{n,m}(k) = 1\}$ denote the NOMA cluster served by UAV m at epoch k. To ensure practical SIC complexity, the cluster size is limited:

$$|\mathcal{N}_m(k)| \le N_m^{\max}, \quad \forall m \in \mathcal{M},\ \forall k \in \mathcal{K}. \tag{11}$$

Let $p_n(k)$ denote the uplink transmit power of WD n at epoch k. Under uplink NOMA, UAV m employs successive interference cancellation (SIC) to decode the superimposed signals. Let $\pi_m(k)$ denote the permutation over $\mathcal{N}_m(k)$ sorted by ascending effective channel gains. The received SINR of WD $n = \pi_m^{(i)}(k)$ at UAV m is

$$\gamma_{n,m}(k) = \frac{p_n(k) g_{n,m}(k)}{\sum_{r=i+1}^{|\mathcal{N}_m(k)|} p_{\pi_m^{(r)}(k)}(k) g_{\pi_m^{(r)}(k),m}(k) + \sigma_0^2}, \quad (12)$$

where σ_0^2 denotes the AWGN noise power. The achievable uplink rate is

$$R_{n,m}(k) = a_{n,m}(k)\, B \log_2\big(1 + \gamma_{n,m}(k)\big), \quad (13)$$

where B denotes the shared uplink bandwidth. The QoS requirement is

$$R_{n,m}(k) \geq a_{n,m}(k)\, R_n^{\min}, \quad \forall n, m, k. \quad (14)$$

The uplink transmit power is bounded by $0 \leq p_n(k) \leq P_n^{\max}$, $\forall n, k$.

D. Task Model with Priority Awareness

1) Task Generation Model: At each epoch $k \in \mathcal{K}$, WD $n \in \mathcal{N}$ generates a task with probability p_0. Let $\delta_n(k) \in \{0, 1\}$ denote the task-generation indicator. A generated task is characterized as

$$T_n(k) = \big(D_n(k),\, F_n(k),\, \tau_n^{\mathrm{dl}}(k),\, \alpha_n(k)\big), \quad (15)$$

where $D_n(k)$ is the input data size (bits), $F_n(k)$ is the required CPU cycles, $\tau_n^{\mathrm{dl}}(k)$ is the task deadline (seconds), and $\alpha_n(k) \in \mathcal{A}_{\mathrm{type}}$ denotes the application type.

2) Composite Task Priority Model: We define a composite priority weight $\omega_n(k)$ for each generated task that captures both deadline-based urgency and application-type importance:

$$\omega_n(k) = \lambda_u \cdot \omega_n^{\mathrm{urg}}(k) + \lambda_t \cdot \omega_n^{\mathrm{type}}(k), \quad (16)$$

where $\lambda_u + \lambda_t = 1$ are weighting coefficients that control the relative emphasis on urgency versus task-type importance.

Deadline-based urgency: The urgency component is inversely proportional to the task deadline, normalized by the maximum deadline in the system:

$$\omega_n^{\mathrm{urg}}(k) = \frac{\tau_{\max}^{\mathrm{dl}} - \tau_n^{\mathrm{dl}}(k)}{\tau_{\max}^{\mathrm{dl}} - \tau_{\min}^{\mathrm{dl}} + \varepsilon}, \quad (17)$$

where $\tau_{\max}^{\mathrm{dl}}$ and $\tau_{\min}^{\mathrm{dl}}$ are the maximum and minimum task deadlines, respectively, and $\varepsilon > 0$ is a small constant for numerical stability. Tasks with shorter deadlines receive higher urgency scores.

Application-type importance: The importance component is determined by the application category of the task:

$$\omega_n^{\mathrm{type}}(k) = w_{\alpha_n(k)}, \quad (18)$$

where $w_\alpha \in (0, 1]$ is a predefined importance weight for application type α. For example, safety-critical applications (e.g., fire detection, collision avoidance) are assigned higher weights than routine monitoring tasks.

Fig. 2 illustrates the composite task priority model.

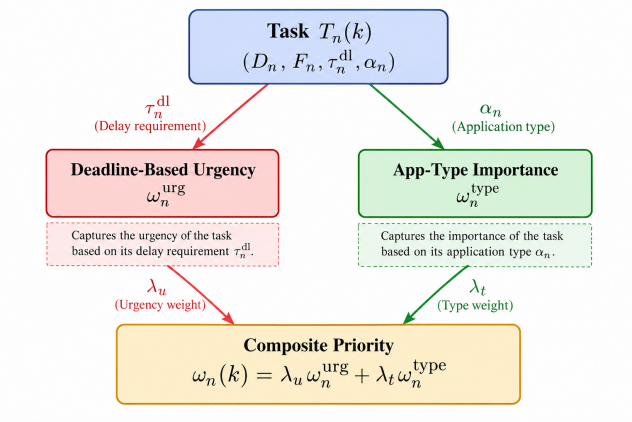


Figure 2. Composite task priority model integrating deadline-based urgency and application-type importance.

E. Computation Model

1) Execution Assignment: Each generated task is executed either locally at its originating WD or remotely at one serving UAV. Let $x_n^{\mathrm{loc}}(k) \in \{0, 1\}$ and $a_{n,m}(k) \in \{0, 1\}$ denote the local execution and UAV offloading indicators. The assignment constraint is

$$x_n^{\mathrm{loc}}(k) + \sum_{m \in \mathcal{M}} a_{n,m}(k) = \delta_n(k), \quad \forall n \in \mathcal{N},\ \forall k \in \mathcal{K}. \quad (19)$$

2) Local Computation Delay: If $x_n^{\mathrm{loc}}(k) = 1$, task $T_n(k)$ is executed at WD n with CPU frequency f_n^{loc}, and the local delay is

$$T_n^{\mathrm{loc}}(k) = \frac{F_n(k)}{f_n^{\mathrm{loc}}}, \quad \forall n \in \mathcal{N},\ \forall k \in \mathcal{K}. \quad (20)$$

3) UAV Offloading Delay: If $a_{n,m}(k) = 1$, task $T_n(k)$ is offloaded to UAV m. The completion delay includes uplink transmission and UAV computation delays. The uplink transmission delay is

$$T_{n,m}^{\mathrm{tx}}(k) = \frac{D_n(k)}{R_{n,m}(k)}. \quad (21)$$

Each UAV maintains a FIFO computation queue. Let $Z_m(k)$ denote the queued workload at UAV m in CPU cycles. Its evolution is

$$Z_m(k+1) = \max\Bigg\{ Z_m(k) + \sum_{n \in \mathcal{N}} a_{n,m}(k) \delta_n(k) F_n(k) - f_m^{\mathrm{uav}} \Delta t,\ 0 \Bigg\}. \quad (22)$$

The corresponding queue waiting delay is

$$Q_m(k) = \frac{Z_m(k)}{f_m^{\mathrm{uav}}}. \quad (23)$$

The UAV computation delay is

$$T_{n,m}^{\mathrm{cmp}}(k) = \frac{F_n(k)}{f_m^{\mathrm{uav}}}. \quad (24)$$

Thus, the total UAV-side task completion delay is

$$T_{n,m}^{\mathrm{uav}}(k) = T_{n,m}^{\mathrm{tx}}(k) + Q_m(k) + T_{n,m}^{\mathrm{cmp}}(k). \quad (25)$$

4) Unified Task Completion Delay: The total task completion delay of WD n at epoch k is

$$T_n^{\mathrm{tot}}(k) = x_n^{\mathrm{loc}}(k) T_n^{\mathrm{loc}}(k) + \sum_{m \in \mathcal{M}} a_{n,m}(k) T_{n,m}^{\mathrm{uav}}(k). \quad (26)$$

5) Priority-Weighted Average Delay: The priority-weighted average task completion delay over all generated tasks is

$$\bar{T}_{\omega} = \frac{\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}} \delta_n(k)\,\omega_n(k)\,T_n^{\mathrm{tot}}(k)}{\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}} \delta_n(k)\,\omega_n(k)}. \tag{27}$$

This metric ensures that delays incurred by high-priority tasks are penalized more heavily than those of low-priority tasks.

F. Energy Consumption Model

The total system energy includes WD local computation, WD uplink transmission, UAV computation, and UAV propulsion energy:

$$E^{\mathrm{total}} = \sum_{n\in\mathcal{N}} \big(E_n^{\mathrm{loc}} + E_n^{\mathrm{tx}}\big) + \sum_{m\in\mathcal{M}} \big(E_m^{\mathrm{cmp}} + E_m^{\mathrm{fly}}\big). \tag{28}$$

1) WD Energy Consumption: Following the dynamic voltage and frequency scaling (DVFS) model, the local computation energy of WD n is

$$E_n^{\mathrm{loc}} = \sum_{k\in\mathcal{K}} x_n^{\mathrm{loc}}(k)\delta_n(k)\kappa_n \big(f_n^{\mathrm{loc}}\big)^2 F_n(k), \tag{29}$$

where κ_n is the effective switched capacitance coefficient. The uplink transmission energy is

$$E_n^{\mathrm{tx}} = \sum_{k\in\mathcal{K}}\sum_{m\in\mathcal{M}} a_{n,m}(k)\delta_n(k)\frac{p_n(k)D_n(k)}{R_{n,m}(k)}. \tag{30}$$

2) UAV Energy Consumption: The onboard computation energy of UAV m is

$$E_m^{\mathrm{cmp}} = \sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}} a_{n,m}(k)\delta_n(k)\kappa_m \big(f_m^{\mathrm{uav}}\big)^2 F_n(k), \tag{31}$$

where κ_m is the UAV capacitance coefficient. For rotary-wing UAVs, the propulsion power at speed $v_m(k)$ is modeled as [10]

$$P_m^{\mathrm{fly}}(k) = \eta_1\left(1+\frac{3v_m^2(k)}{v_{\mathrm{tip}}^2}\right) + \eta_2\left(\sqrt{\eta_3+\frac{v_m^4(k)}{4}}-\frac{v_m^2(k)}{2}\right)^{\frac{1}{2}} + \eta_4 v_m^3(k), \tag{32}$$

where v_{tip} and $\eta_1,\eta_2,\eta_3,\eta_4$ are aerodynamic constants, and $v_m(k) = \frac{\|\mathbf{q}_m(k+1)-\mathbf{q}_m(k)\|_2}{\Delta t}$. The total propulsion energy is

$$E_m^{\mathrm{fly}} = \Delta t \sum_{k=1}^{K-1} P_m^{\mathrm{fly}}(k). \tag{33}$$

The UAV energy budget constraint is

$$E_m^{\mathrm{cmp}} + E_m^{\mathrm{fly}} \le E_m^{\max}, \quad \forall m\in\mathcal{M}. \tag{34}$$

G. Problem Formulation

The objective is to minimize the priority-weighted average task completion latency while regularizing UAV energy consumption, through joint optimization of UAV 3D trajectory, priority-aware task offloading, and NOMA transmission power decisions. Due to binary offloading variables, continuous trajectory/power variables, and nonlinear coupling among communication, computation, and energy models, the problem is formulated as a stochastic MINLP:

$$\min_{X} \ \frac{\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}} \delta_n(k)\,\omega_n(k)\,T_n^{\mathrm{tot}}(k)}{\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}} \delta_n(k)\,\omega_n(k)} + \mu\sum_{k\in\mathcal{K}}\sum_{m\in\mathcal{M}} E_m^{\mathrm{total}}(k) \tag{35}$$

where $\mu \ge 0$ is a weighting coefficient controlling the latency-energy trade-off, $X = \{\mathbf{q}_m(k),\, a_{n,m}(k),\, x_n^{\mathrm{loc}}(k),\, p_n(k)\}$ is the set of optimization variables, and $E_m^{\mathrm{total}}(k) = E_m^{\mathrm{cmp}}(k) + E_m^{\mathrm{fly}}(k)$.

Subject to:

$$\mathcal{C}_1: \ x_n^{\mathrm{loc}}(k) + \sum_{m\in\mathcal{M}} a_{n,m}(k) = \delta_n(k), \quad \forall n,k \tag{36}$$

$$\mathcal{C}_2: \ 0 \le p_n(k) \le P_n^{\max}, \quad \forall n,k \tag{37}$$

$$\mathcal{C}_3: \ \|\mathbf{q}_m(k+1)-\mathbf{q}_m(k)\| \le v_m^{\max}\Delta t, \quad \forall m,k \tag{38}$$

$$\mathcal{C}_4: \ h_{\min} \le h_m(k) \le h_{\max}, \quad \forall m,k \tag{39}$$

$$\mathcal{C}_5: \ E_m^{\mathrm{cmp}} + E_m^{\mathrm{fly}} \le E_m^{\max}, \quad \forall m \tag{40}$$

$$\mathcal{C}_6: \ \mathbf{q}_m(k) \in \mathcal{A}, \quad \forall m,k \tag{41}$$

$$\mathcal{C}_7: \ |\mathcal{N}_m(k)| \le N_m^{\max}, \quad \forall m,k \tag{42}$$

$$\mathcal{C}_8: \ R_{n,m}(k) \ge a_{n,m}(k)\,R_n^{\min}, \quad \forall n,m,k \tag{43}$$

$$\mathcal{C}_9: \ T_n^{\mathrm{tot}}(k) \le \tau_n^{\mathrm{dl}}(k), \quad \forall n,k \tag{44}$$

Constraint $\mathcal{C}_1$ enforces unique task assignment. $\mathcal{C}_2$ imposes transmit power limits. Constraints $\mathcal{C}_3$–$\mathcal{C}_6$ define UAV mobility, altitude, energy budget, and deployment-region constraints. $\mathcal{C}_7$ and $\mathcal{C}_8$ enforce NOMA cluster-size and QoS requirements, respectively. Constraint $\mathcal{C}_9$ ensures that task completion does not exceed the deadline.

Remark: The problem (35)–(44) is a stochastic MINLP due to binary offloading and assignment decisions, continuous trajectory and power decisions, nonlinear SINR and rate functions, and stochastic task arrivals and WD mobility. This class of problems is NP-hard in general, motivating the use of multi-agent deep reinforcement learning for tractable approximate solution.

IV. Proposed Methodology

[This section will present the MAPPO-based solution framework, including the MDP formulation, state/action/reward design, CTDE architecture, NOMA bisection power allocation, and the joint optimization algorithm. To be completed in the next phase.]

V. Simulation Results

[This section will present simulation setup, benchmark comparison, convergence analysis, trajectory visualization, and performance evaluation under varying network conditions. To be completed after implementation.]

VI. Conclusion and Future Work

[This section will summarize the contributions, key findings, and outline directions for future work including online adaptation and real-world deployment. To be completed.]

REFERENCES

[1] Y. Yin *et al.*, "Integrated Sensing and Communication for UAV Networks," *IEEE Transactions on Wireless Communications*, vol. 24, no. 3, pp. 1812–1827, 2025.

[2] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A Survey on Mobile Edge Computing: The Communication Perspective," *IEEE Communications Surveys and Tutorials*, vol. 19, no. 4, pp. 2322–2358, 2017.

[3] X. Liu, J. Zhang, and W. Chen, "NOMA-Enhanced UAV MEC Networks," *IEEE Internet of Things Journal*, vol. 11, no. 5, pp. 4521–4535, 2024.

[4] H. Kang, X. Chang, J. Mišić, V. B. Mišić, J. Fan, and Y. Liu, "Cooperative UAV Resource Allocation and Task Offloading in Hierarchical Aerial Computing Systems: A MAPPO-Based Approach," *IEEE Internet of Things Journal*, vol. 10, no. 12, pp. 10 497–10 509, 2023.

[5] Y. Zeng, J. Lyu, and R. Zhang, "Cellular-Connected UAV: Potential, Challenges, and Promising Technologies," *Proceedings of the IEEE*, vol. 107, no. 2, pp. 2322–2358, 2019.

[6] L. Yang, J. Zheng, and Y. Zhang, "A Deep Reinforcement Learning-Based Service Caching and Computation Offloading Algorithm for Multi-UAV-Enabled MEC Systems," *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 6658–6669, 2026.

[7] M. N. Tariq, J. Wang, S. Raza, M. Siraj, M. Altamimi, and S. Memon, "Toward Optimal Resource Allocation: A Multi-Agent DRL Based Task Offloading Approach in Multi-UAV-Assisted MEC Networks," *IEEE Access*, vol. 12, pp. 81 428–81 440, 2024.

[8] D. C. Barman, S. Chakraborty, A. Hazra, and N. Mazumdar, "Joint Service Caching, Task Offloading, and Service Delivery in Mobile Edge Computing for Latency Critical UAV Networks," *IEEE Transactions on Computational Social Systems*, pp. 1–13, 2026.

[9] Y. Zeng, R. Zhang, and T. J. Lim, "Throughput Maximization for UAV-Enabled Mobile Relaying Systems," *IEEE Transactions on Communications*, vol. 64, no. 12, pp. 4983–4996, 2016.

[10] Y. Zeng, J. Xu, and R. Zhang, "Energy Minimization for Wireless Communication with Rotary-Wing UAV," *IEEE Transactions on Wireless Communications*, vol. 18, no. 4, pp. 2329–2345, 2019.

[11] N. C. Luong *et al.*, "Applications of Deep Reinforcement Learning in Communications and Networking," *IEEE Communications Surveys and Tutorials*, vol. 21, no. 4, pp. 3133–3174, 2019.

[12] D. Nguyen *et al.*, "Dependency-Aware Task Offloading in Collaborative UAV-MEC Networks," *IEEE Transactions on Mobile Computing*, vol. 22, no. 11, pp. 6521–6535, 2023.

[13] Z. Hu, F. Zheng, S. Long, and H. Zhu, "Priority-Aware Task Offloading for UAV-Assisted Mobile Edge Computing," *IEEE Access*, vol. 8, pp. 189 534–189 545, 2020.

[14] W. Sun, J. Yang, Y. Zhang, and S. Liu, "Priority-Based Task Offloading and Resource Allocation in UAV-Assisted MEC," *IEEE Internet of Things Journal*, vol. 10, no. 15, pp. 13 485–13 498, 2023.

[15] H. Peng, F. Wu, and X. Chen, "Priority-Aware Multi-UAV Trajectory and Task Scheduling in MEC Systems," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, pp. 8234–8248, 2024.

[16] Y. Sun *et al.*, "Two-Timescale Joint Communication and Computation for UAV-Enabled MEC," *IEEE Transactions on Communications*, vol. 71, no. 6, pp. 3511–3527, 2023.

[17] C. Zhan, W. Liu, K. Song, R. Fan, J. Liu, and H. Hu, "Joint UAV Placement and Dependent Task Offloading in Multi-UAV MEC Networks," *IEEE Transactions on Mobile Computing*, vol. 25, no. 4, pp. 5285–5301, 2026.

[18] Z. Qin *et al.*, "DRL-Based Joint Trajectory and NOMA Resource Allocation for UAV-Assisted MEC," *IEEE Transactions on Communications*, vol. 72, no. 3, pp. 1834–1849, 2024.

[19] T. Camp, J. Boleng, and V. Davies, "A Survey of Mobility Models for Ad Hoc Network Research," *Wireless Communications and Mobile Computing*, vol. 2, no. 5, pp. 483–502, 2002.